DUAL-Compass: Direction-Aware Simultaneous Learning and Unlearning for Explainable VLM Hallucination Mitigation

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Abstract

Hallucination in vision–language models (VLMs) is conventionally treated as an unsigned scalar to be minimized, so mitigation methods apply a single corrective pressure, suppressing over-claims, uniformly across models and training runs. We argue instead that hallucination is a *signed* phenomenon: the same model pool contains failures of over-claiming (asserting absent objects; *yes-bias*) and failures of over-withholding (missing present objects; *no-bias*), and the two require *opposite* parameter updates. We present **DUAL-Compass**, a training framework that elevates an explainable, direction-aware hallucination diagnosis into the learning signal itself. The diagnosis partitions a model's own errors into an *unlearn* stream (overclaims, suppressed with negative-preference optimization), a *learn* stream (missed objects, reinforced with supervised updates), and a *retain* stream (faithful anchors), which are optimized *simultaneously* under a single objective. A compass-guided closed loop periodically re-diagnoses the model during training and re-balances the two directional pressures until the signed error score converges to zero. Across three VLMs (2B to 7B, two architectural families), compass-guided training with balance checkpointing improves held-out *adversarial* accuracy on every model with zero collapses across nine loop runs (three seeds and three models), reducing the dominant error direction toward balance (on Qwen2.5-VL-3B the signed score moves from +0.76 to −0.07 and accuracy from 83.2% to 88.8%; the reduction is partial on the 2B model, which we trace to a probe-distribution effect). A controlled stress test isolates the mechanism: under fixed weights both negative-preference and gradient-ascent unlearning collapse when pushed, whereas under closed-loop control both are stable. The collapse is therefore a property of missing feedback rather than of the loss function, and the diagnosis-driven controller is the contribution. Every alternative fails in a characteristic way: fixed-weight simultaneous training is fragile to data ordering and collapses on two of three models; balanced supervised fine-tuning collapses toward universal denial (60.3% ± 14.4 accuracy over three seeds); unlearning alone leaves behavior essentially unchanged; learning alone collapses the model to universal assertion (100% false-positive rate); and an EFUF-style suppression-only baseline matches peak accuracy yet overshoots into no-bias (signed −0.67 against our +0.53) while remaining budget-fragile. We further contribute a direction-balanced data-expansion protocol that doubles the supervision pool from per-image bias-target annotations, and document a cautionary measurement result: small-sample error counts can invert the diagnosed bias direction on five of six panel models, motivating diagnosis-driven rather than count-driven intervention. The standard external POPE benchmark (COCO) independently confirms this direction-dependence: the loop repairs the one model that is yes-biased there (bias from +0.41 to +0.15, accuracy up) and is direction-mismatched on the two whose COCO direction is the opposite (flat on 3B, mildly worse on 2B). Re-diagnosing those models *on* COCO reverses the mismatch and improves the flagship 2B from 73.0% to 91.3% on held-out POPE-COCO, direct evidence that repair follows direction. A blind human audit (three annotators) confirms both halves of the method: annotators agree with the diagnosed direction on 90% of items and prefer the repaired model's captions over the inference-time and EFUF-style baselines on 85%. Code, data, and trained adapters are available (anonymized for review) at https://anonymous.4open.science/r/DUAL-Compass-DD12.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Information systems** → *Data mining*.

Keywords

vision–language models, hallucination, machine unlearning, trustworthy machine learning, explainable diagnosis, preference optimization

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1 Introduction

Large vision–language models (VLMs) hallucinate: they describe objects that are not in the image, and they overlook objects that are [11, 13, 16, 24]. As VLMs move into deployment, assistive description for blind users, content moderation, robotics, medical triage, both failure directions carry real cost, though of different kinds: asserting what is absent makes a system *unsafe* (false alarms, fabricated evidence), while denying what is present makes it *unhelpful* (missed objects, incomplete descriptions). A rapidly growing literature measures hallucination [4, 8, 19, 25, 27, 28, 33] and mitigates it [9, 12, 14, 15, 34–38, 40]. Almost all of this work shares one structural assumption: hallucination is a *magnitude*, a rate to be pushed down. The corrective pressure is therefore one-directional, typically suppressing the model's tendency to over-claim.

Recent diagnostic work challenges this assumption. HalluCompass [2] showed that aggregate metrics such as POPE's yes-ratio

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arithmetically cancel two qualitatively different failure modes, *yes-bias* (over-claiming absent objects) and *no-bias* (withholding supported evidence), and that prompt-level interventions chosen against the diagnosed direction are strictly harmful: applying a chain-of-thought prompt to a yes-biased model raised its false-positive rate by up to 23.3 percentage points. Related observations of sycophantic assent bias in language models [26] reinforce the picture that assertion bias is a directional, model-specific property. However, that line of work stops at *evaluation*: the diagnosis selects a prompt, not a parameter update. Prompt-mode routing is superficial in the sense that the model's weights, and thus its underlying propensity to hallucinate, are unchanged; the intervention must be re-applied at every inference call and can be silently dropped by downstream users.

On the training side, machine unlearning, originally developed for privacy-motivated data removal [5, 6, 31], has recently been applied to hallucination mitigation. EFUF [30] uses CLIP similarity to locate hallucinated spans in captions and applies gradient ascent to unlearn them; follow-up work moves the unlearning target into the vision encoder to combat superficial forgetting [3]. Yet these methods inherit the unsigned view in three ways. First, the signal that selects *what* to unlearn is an opaque scalar (a CLIP similarity threshold): it cannot state what *kind* of failure is being removed. Second, only the over-claiming direction is treated; nothing in the objective *teaches* the model the objects it currently misses. Third, the mixing between forgetting and retention is fixed by hand. General-purpose unlearning research meanwhile shows that forgetting and retaining need not conflict, they can be optimized simultaneously with aligned gradients [1, 7, 32, 39], but in that literature the forget set is externally given (privacy, copyright), not produced by a diagnosis of the model's own errors.

This paper closes the gap between the two lines. We propose **DUAL-Compass** (Direction-aware Unlearning-And-Learning), a training framework in which an explainable, signed hallucination diagnosis becomes the supervision itself. Our contributions are:

- **C1, Diagnosis as supervision.** The model's own errors on a presence-probing pool are split by direction into an *unlearn* stream (over-claims), a *learn* stream (missed objects), and a *retain* stream (correct behavior). Because every training example carries its diagnostic direction, the intervention is auditable at the level of individual failure types: one can state which direction of error was suppressed, which was reinforced, and why the balance shifted during training (Section 4.1).
- **C2, A simultaneous, direction-symmetric objective.** A single three-term loss applies *opposite gradient pressures at the same time*: negative-preference optimization (NPO) [39] suppresses hallucinated assertions, supervised learning reinforces missed detections, and a retain term anchors utility (Section 4.2). Our ablations show this simultaneity is a *requirement*, not a convenience: each single-direction pressure in isolation is either ineffective or actively harmful (Section 5.2).
- **C3, Compass-guided closed-loop control.** Fixed mixing weights overshoot the balance point, converting a yes-biased model into a no-biased one. DUAL-Compass treats the signed score as a control variable: the model is re-diagnosed every K steps, the stream-sampling probabilities are re-set from the current score, and training terminates when the score converges to zero (Section 4.3).
- **C4, Direction-balanced data expansion and a measurement caution.** We mine per-image bias-target annotations into new presence probes, doubling supervision to 8,006 queries while growing *both* directional streams (Section 4.4). We also document that small-sample error counts invert the diagnosed bias direction on five of six panel models relative to a judge-based diagnosis, so count-driven routing on small samples can push a model the wrong way, our learn-only ablation shows exactly how damaging that is (Section 5.9).

Figure 1 summarizes the framework.

Findings. With LoRA adapters [10] on Qwen2-VL-2B, Qwen2.5-VL-3B [29], and LLaVA-1.5-7B [18], the experimental picture is consistent and sharp. (i) The best fixed-weight DUAL run reduces the held-out over-claim rate from 80.9% to 5.5% while raising accuracy from 64.0% to 86.0%, but fixed weights are *fragile*: across seeds and models they range from this best case (82.4% ± 4.6 over three 2B seeds) down to outright collapse (an expanded-pool seed at 44.4%; collapse on 3B and 7B). (ii) Single-direction interventions fail symmetrically: unlearning alone is inert (+0.8pp), learning alone collapses to universal assertion (100% FP), a training-level reproduction of the finding that direction-misaligned interventions are strictly harmful [2], and balanced supervised fine-tuning, the natural "simple" alternative, collapses toward universal denial (60.3% ± 14.4). (iii) The compass-guided closed loop with balance checkpointing repairs all of this: on all three models it improves accuracy (86.4%/88.8%/84.8%) and reduces the dominant error direction with zero collapses. The controller is not an optimization nicety; it is what converts a fragile objective into a reliable procedure.

2 Related Work

2.1 Measuring VLM Hallucination

CHAIR [24] measures hallucinated objects in generated captions via object-level recall against COCO segmentation. POPE [16] recast object hallucination as binary presence probing with random, popular, and adversarial negative-object splits, and introduced the yes-ratio as a response-bias indicator; NOPE [19] extended negative-presence evaluation. AMBER [28] added attribute and relation hallucination with an LLM-free pipeline; HallusionBench [8] targets entangled language hallucination and visual illusion; MMHal-Bench [27] evaluates free-form response quality. BEAF [33] manipulates the visual scene itself (before/after object removal) and scores answer *changes*, introducing direction-adjacent metrics such as stubbornness and indecision. HALLUCINOGEN [25] showed that implicit prompts (localization, visual-context, counterfactual) elicit substantially more hallucination than explicit probes, and DASH [4] mined systematic false-positive clusters from web-scale data, finding more than 19k clusters across 950k images. HalluCompass [2] unified these threads into a signed, direction-aware diagnosis, separating yes-bias from no-bias with an adjective-based judge protocol, and a routing rule over prompt modes. All of these are evaluation-only; none feeds the diagnosis back into training. DUAL-Compass is the training-side completion of this program.

2.2 Mitigating VLM Hallucination

Mitigation approaches fall into four groups. *Decoding-time* methods reshape token distributions at inference: VCD [15] contrasts against distorted visual input, OPERA [12] penalizes over-trust attention patterns, and EOS-decision work [38] shows that simply ending generation earlier reduces hallucinated tails. *Post-hoc* methods revise generated text: LURE [40] trains a revisor targeting co-occurrence, uncertainty, and position factors; Woodpecker [34] corrects hallucinations through a five-stage validation pipeline. *Data-side* methods clean or augment training corpora: HalluciDoctor [35] removes hallucinatory instruction data, and HACL [14] adds hallucination-augmented contrastive pairs. *Feedback-based* methods fine-tune against preference signals: M-HalDetect [9] trains reward models on fine-grained hallucination annotations, RLHF-V [36] uses segment-level human corrections, RLAIF-V [37] replaces human feedback with open-source AI feedback, and factually-augmented RLHF [27] enriches the reward model with ground-truth facts. All four groups treat hallucination as unsigned: the corrective pressure is uniformly anti-over-claim. Our ablation shows why this is risky, the same pressure that helps a yes-biased model is precisely the pressure a no-biased model does not need, and the opposite single pressure (pro-assertion) collapses the model (Section 5.2).

2.3 Machine Unlearning

Machine unlearning originates in privacy and the right to be forgotten: exact unlearning by system design [6] and sharded retraining (SISA) [5], surveyed in [31]. For LLMs, approximate unlearning by gradient ascent is prone to catastrophic collapse; NPO [39] reframes unlearning as preference optimization on negative samples only and provably diverges exponentially slower, making it the standard strong baseline; efficient variants add lightweight unlearning layers [7] and task vectors [32]. TOFU [20] established evaluation protocols for forget quality versus model utility. LUR [1] shows forgetting and retention can be optimized simultaneously by aligning their gradients in a bi-level scheme. In the multimodal setting, EFUF [30] unlearns CLIP-flagged hallucinated spans with three tailored losses, and encoder-focused reinforcement unlearning [3] addresses superficial forgetting where the language decoder avoids a token while visual representations persist. Distinct from all of these, in DUAL-Compass (i) the forget and learn sets are not externally given but produced by diagnosing the model's own errors; (ii) the two sets receive *opposite* pressures simultaneously; and (iii) the balance between pressures is closed-loop controlled by an explainable signed score.

2.4 Model Editing and Preference Optimization

Knowledge-editing methods such as ROME [21] and MEMIT [22] rewrite individual factual associations by locating and editing specific weights; they share our goal of targeted behavioral repair but operate on declarative facts rather than directional response biases, and offer no mechanism for balancing two opposing behavioral tendencies. Direct preference optimization [23] and its negative-only variant NPO [39] provide the reference-anchored loss family we build on; our contribution is not a new loss but a new *supervision structure*, direction-partitioned streams under closed-loop balance control, into which these losses slot.

2.5 Positioning

Table 8 summarizes how DUAL-Compass differs from representative mitigation and unlearning approaches along five properties: whether the selection signal is explainable (states *what kind* of failure is treated), whether both error directions are addressed, whether learning and unlearning are simultaneous, whether the balance is closed-loop controlled, and whether the fix is permanent (weight-level rather than per-inference).

3 Preliminaries and Problem Setup

Data. We build on the HalluCompass evaluation suite [2]: 2,200 images spanning curated (COCO, AMBER) and in-the-wild (NoCaps, VizWiz) sources. Each image carries annotations of present objects (`truth`), plausible absent objects mined to be tempting over-claims (`hallu_pop`, `hallu_adv`), and per-image direction targets (`yes_bias_targets`: absent objects the model family tends to assert; `no_bias_targets`: present objects it tends to miss). The suite releases 4,006 binary presence queries ("*Is there a ⟨object⟩ in this image?*") in POPE-style random/popular/adversarial splits. Full prompt and probe templates appear in Appendix B.

Notation is summarized in Appendix N.

Signed diagnosis. For a model π_θ evaluated on a probe pool, let FP be over-claims (label *no*, predicted *yes*) and FN be misses (label *yes*, predicted *no*). The *behavioral signed score* is

$$s = \frac{\mathrm{FP} - \mathrm{FN}}{\mathrm{FP} + \mathrm{FN}} \in [-1, +1], \tag{1}$$

where $s > 0$ indicates yes-bias (over-claiming dominates) and $s < 0$ no-bias (over-withholding dominates). HalluCompass additionally provides an adjective-based judge diagnosis (DSA): a judge LLM converts each wrong prediction into failure-describing adjectives, which a frozen 16-word partition maps to yes-bias, no-bias, or generic classes; the normalized difference yields a judge-side signed score. On our primary subject model Qwen2-VL-2B the published DSA score is +0.252, the strongest yes-bias in the ten-model panel, and our full-pool behavioral score agrees in sign ($s = +0.386$; Section 5.9).

Problem. Given a pretrained VLM π_θ and a presence-probe pool with ground truth, produce updated parameters θ' such that (i) the over-claim rate drops, (ii) the miss rate does not inflate to compensate, (iii) the signed score converges toward zero, and (iv) behavior on already-correct inputs is preserved, using only parameter-efficient updates and the model's own diagnosed errors as supervision.

4 The DUAL-Compass Framework

4.1 Diagnosis-to-Streams Construction

Running the subject model once over the presence-probe pool partitions every query into exactly one stream:

- **Unlearn stream $\mathcal{U}$** (yes-bias evidence): the object is absent, the model asserted presence. The assertion ("Yes.") is the behavior to remove.

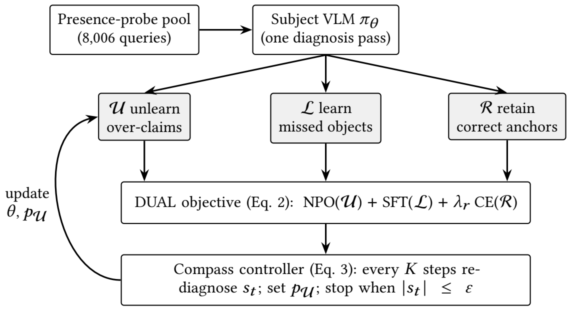

Figure 1: DUAL-Compass. A single diagnosis pass splits the model's own errors by direction into unlearn/learn/retain streams; the three streams are optimized simultaneously; a closed-loop controller re-diagnoses the signed score s_t during training and stops at directional balance.

- **Learn stream $\mathcal{L}$** (no-bias evidence): the object is present, the model denied it. The correct assertion is the behavior to instill.
- **Retain stream $\mathcal{R}$**: correct predictions, kept as anchors against drift and collapse.

Each record carries its image, query, gold answer, and diagnosed direction. This makes the downstream update auditable per failure type: after training one can report, per direction, how many diagnosed failures were repaired, how many persisted, and how many new failures the update introduced. For our primary subject model, a single inference pass over the 4,006-query pool yields $|\mathcal{U}| = 289$, $|\mathcal{L}| = 128$, and $|\mathcal{R}| = 3{,}589$ (Table 9), constructing the streams requires one forward pass per query and no human labeling. Over-claims outnumber misses in every split and peak on the adversarial split (119 vs. 44), consistent with the model's published yes-bias diagnosis; the stream construction thus reproduces the direction profile that the judge-based diagnosis reports, from behavior alone. Per-source breakdowns, which reveal a source-dependent direction gradient, appear in Appendix D.

4.2 Simultaneous Objective

Let y^+ denote the gold answer for a learn/retain record and y^- the hallucinated assertion ("Yes.") for an unlearn record. With reference model π_{ref} (the frozen base; implemented at zero memory cost by disabling the LoRA adapter), the DUAL objective over a mini-batch is

$$\mathcal{L}_{\text{DUAL}} = \underbrace{\mathbb{E}_{\mathcal{L}}\left[-\log \pi_\theta(y^+ \mid v, x)\right]}_{\text{learn (no-bias)}} + \lambda_r \underbrace{\mathbb{E}_{\mathcal{R}}\left[-\log \pi_\theta(y^+ \mid v, x)\right]}_{\text{retain}} + \lambda_u \underbrace{\mathbb{E}_{\mathcal{U}}\left[-\frac{2}{\beta}\log \sigma\left(-\beta \log \frac{\pi_\theta(y^- \mid v,x)}{\pi_{\text{ref}}(y^- \mid v,x)}\right)\right]}_{\text{unlearn (yes-bias), NPO}}. \tag{2}$$

The unlearn term is NPO [39] rather than naive gradient ascent, inheriting its exponentially slower divergence from the reference policy. Two design choices matter. First, relative to EFUF [30], the positive term is not merely a fluency regularizer: it is an explicit *directional* objective on the model's diagnosed no-bias failures, so the objective pushes the two error directions toward zero *at the same time*. Second, the reference-anchored form of the unlearn term means suppression is measured against the base model, not against zero probability, the update lowers the assertion's likelihood ratio rather than chasing degenerate outputs, which is what makes simultaneous optimization with the learn term stable. A gradient-level analysis of both properties, and of why the learn term alone collapses, appears in Appendix M.

4.3 Compass-Guided Closed Loop

Fixed mixing weights overshoot. As Section 5.2 shows, constant-pressure DUAL training drives the signed score from +0.978 through zero to −0.657: it converts a yes-biased model into a no-biased one. The corrective pressures are effective but lack a termination criterion.

DUAL-Compass therefore treats the signed score as a control variable. Every K steps, the model is re-diagnosed on a held-out balanced probe set to obtain s_t, and the probability of drawing an unlearn example is set to

$$p_{\mathcal{U}}(t) = \text{clip}\left(0.5 + \gamma\, s_t,\ p_{\min},\ p_{\max}\right), \tag{3}$$

with the remaining mass allocated to learn and retain draws. Training terminates early once $|s_t| \le \varepsilon$ on consecutive probes, i.e., when the diagnosis reports directional balance. In addition, every probe records a *best-balance checkpoint*, the adapter state with the smallest $|s_t|$ so far, ties broken by probe accuracy, and the final model restores this checkpoint, so a run that passes through balance cannot end in a worse phase. Table 17 summarizes the procedure. We report an undamped configuration (γ=0.6, lr 10^{-4}, K=25) and a damped one (γ=0.3, lr 3×10^{-5}, K=20); $p_{\min}$=0.15, $p_{\max}$=0.85, ε=0.1 throughout. The controller has a natural interpretation: γ is a proportional gain mapping diagnosed imbalance to corrective pressure, and ε defines the dead-band within which the model is considered directionally balanced.

4.4 Direction-Balanced Data Expansion

Presence-probe supervision is bounded by the released query set. We expand it directly from the per-image annotations: every annotated plausible-absent object (`hallu_pop` ∪ `hallu_adv`, excluding objects listed in `truth`) yields a new absent probe, and every annotated present object yields a new present probe. The annotation pool admits 14,439 absent and 9,961 present candidates; after deduplication against the released queries (3,816 unique image–object pairs) we sample a balanced pool of 2,000 absent + 2,000 present probes, doubling total supervision to 8,006 queries. Because the expansion is derived from bias-target annotations, both directional streams grow, not just the over-claim side, a property no existing hallucination-mitigation corpus provides, since preference-pair datasets [9, 36] encode only which response is *less hallucinated*, never which *direction* the error runs. Generation rules, candidate-pool statistics, and per-source composition of the expansion appear in Appendix C.

4.5 Cost

DUAL-Compass adds three costs on top of standard LoRA fine-tuning: (i) one diagnosis pass over the probe pool (a single forward per query; ~25 minutes for 8k queries on one H100 for a 2B model); (ii) one probe evaluation per K steps ($|\mathcal{P}|$ = 80 forward pairs); and (iii) one reference forward per unlearn step (adapter-disabled, no extra memory). No human labels, judge API calls, or reward-model training are required; the entire pilot fits in a few GPU-hours. For calibration: EFUF reports roughly 3 GPU-hours against 20+ for RLHF-based alignment on the same scale [30]; DUAL-Compass sits in the same low-cost regime while additionally producing the per-direction audit and requiring no CLIP scoring pass, and is far cheaper than preference-data pipelines that require human or proprietary-model feedback [27, 36].

5 Experiments

We organize the evaluation around four research questions:

- **RQ1.** Does simultaneous direction-aware training outperform single-direction unlearning or learning? (Sec. 5.2)
- **RQ2.** Does closed-loop control prevent the overshoot of fixed-weight training? (Sec. 5.4)
- **RQ3.** Does binary rebalancing preserve free-form generative utility? (Sec. 5.5)
- **RQ4.** Do the findings generalize across models, and how sensitive is the diagnosis to measurement regime? (Sec. 5.9)

In one line: every alternative to compass-guided control fails in a characteristic, reproducible way, and control succeeds on every model; a claim-by-claim summary of the results below is in Appendix W (Table 30).

5.1 Setup

Subject models. Qwen2-VL-2B-Instruct [29] is the primary subject: it carries the strongest published yes-bias diagnosis in the HalluCompass panel (DSA +0.252), making it the clearest test of directional repair. Qwen2.5-VL-3B and LLaVA-1.5-7B [17, 18] (the latter a different architectural family, published DSA +0.172, high-FP profile) serve as generalization targets.

Training. LoRA [10] (r=16, α=32, dropout 0.05) on language-tower attention projections only; the vision encoder is frozen. AdamW, learning rate 10^{-4}; NPO β=0.1; 150 steps for fixed-weight ablations; up to 300 steps for the closed loop. Streams are built from the random/popular (+expanded) splits; *the adversarial split is held out entirely* from all training and probing.

Evaluation. On the held-out adversarial split we score the model by comparing the total log-likelihood of "Yes." vs. "No." given the image and question (we report on a 250-query subset for the ablation and cross-model tables (Tables 1, 3, 2, 4) and additionally on the *full* 1,341-query split for the prompting comparison (Table 12); on the trained models the two regimes agree closely (2B loop 86.4% subset vs. 86.8% full), while the base rate differs by ~2pp (64.0% vs. 61.7%) because the balanced subset over-represents the adversarial absent objects), then report FP rate (over-claims among absent-object queries), FN rate (misses among present-object queries), accuracy, and the signed score of Eq. (1). The log-likelihood probe is deliberately more sensitive than greedy decoding: it exposes the model's latent assertion preference even where sampled text would hedge, surfacing an 80.9% latent FP rate where greedy decoding shows

Table 1: Direction-aware training on Qwen2-VL-2B, evaluated on 250 held-out adversarial queries (log-likelihood probe). Only the simultaneous objective improves both directions of failure and accuracy. A representative seed is shown; over three seeds held-out accuracy is 64.8/57.2/61.6 (unlearn-only, 61.2±3.8), 56.0/56.0/56.0 (learn-only, 56.0±0.0), and 86.0/77.2/84.0 (DUAL fixed, 82.4±4.6), the ordering is seed-invariant. s is the behavioral signed score (Eq. 1).

	FP rate	FN rate	Acc.	s
Base model	80.9%	0.7%	64.0%	+0.978
Unlearn only (NPO)	79.1%	0.7%	64.8%	+0.977
Learn only (SFT)	100.0%	0.0%	56.0%	+1.000
DUAL (fixed weights)	**5.5%**	20.7%	**86.0%**	−0.657

14.3% on the same pool. Conclusions about *relative* ablation behavior are invariant to the probe choice; we report both regimes for transparency.

Implementation. All training runs use Unsloth-optimized model patching over HuggingFace Transformers on H100 GPUs; per-run wall-clock for a 150-step ablation including both evaluations is under one hour for the 2B model. Table 21 lists all hyperparameters; none were tuned per ablation arm, all arms share identical settings, so differences in Table 1 are attributable to the objective alone. Evaluation is deterministic (log-likelihood comparison, no sampling); the main comparison arms of Table 1 and the balanced-SFT baseline are reported over three seeds, and a representative seed is shown in each per-run table. Full environment details are given in Appendix I, and an artifact manifest with exact commands in Appendix J.

5.2 RQ1: Only Simultaneous Training Works

Table 1 contains the central evidence from the controlled single-order comparison; multi-seed statistics follow in observation (3) and Section 5.4. Three observations:

(1) The simultaneous objective is the only one that works. DUAL training removes 75.4pp of over-claim rate and adds 22pp of accuracy in 150 LoRA steps, using only the model's own diagnosed errors as supervision. Neither component achieves this alone, at the same step budget, learning rate, and data pool.

(2) Single-direction interventions fail in two distinct ways. *Unlearning alone produces almost no behavioral change*: NPO's reference-anchored suppression, without a counterweighting assertion target, leaves the yes-bias intact (s: +0.978 to +0.977; all three seeds stay strongly yes-biased, $s \geq +0.98$). This is consistent with NPO's design objective of slow, stable forgetting, and shows that stable forgetting alone cannot rebalance a heavily skewed assertion prior. *Learning alone is actively harmful*: supervising only the missed-"Yes" cases amplifies the model's existing yes-bias until it answers "yes" universally (FP 100%, accuracy −8pp, identical to the tenth of a point across all three seeds, 56.0±0.0). This is a training-level reproduction of the evaluation-level finding that interventions misaligned with the diagnosed direction can be strictly harmful [2]: the learn stream is precisely the appropriate intervention for a no-biased model and precisely the harmful one for this yes-biased model, unless it is applied jointly with the opposing pressure. The risk is not hypothetical: a practitioner who observed only the model's misses (e.g., user complaints about undetected objects) and fine-tuned on them would produce precisely this collapse.

(3) Fixed weights overshoot, and are fragile. Even the successful DUAL run does not stop at balance: the signed score crosses zero and lands at −0.657 (FN rises to 20.7%). Worse, the degree of over-correction depends heavily on seed and data ordering: over three seeds fixed-weight DUAL spans 77.2%–86.0% accuracy (82.4 ± 4.6) on the 2B pool, on the expanded pool one of three seeds collapses to 44.4%, and on Qwen2.5-VL-3B and LLaVA-1.5-7B the same fixed recipe collapses outright (44.4% and 58.4%; Table 3). The corrective pressures work, yet without feedback the outcome depends on where training happens to stop. This converts the signed diagnosis from a post-hoc report into the missing control signal (Section 4.3).

Qualitative stream examples. The diagnosed over-claims are contextually plausible absences (a saddle near a horse, a coaster near a cup, an air conditioner in a room photo), the co-occurrence-driven assertions decoding-time methods target statistically but cannot attribute, while the misses include even high-salience objects (a television, the sky), so no-bias is not confined to small or peripheral objects. Representative and extended direction-tagged records across all four sources appear in Appendix K.

5.3 The Collapse Is About Feedback, Not the Loss Function

A natural objection to our use of NPO is that a simpler gradient-ascent (GA) unlearning term, as in EFUF [30], might do as well. At the 150-step budget of Table 1 it does (GA+retain reaches 85.6% vs. DUAL's 86.0%). We resolve this apparent tie with a two-axis stress test (Table 2), and the answer reframes the contribution.

Under fixed weights, both losses are fragile. Pushing the budget to 450 steps, NPO degrades less than GA (signed −0.75 vs. −0.97), the exponentially-slower-collapse property NPO is designed for. Pushing *intensity* instead, however, collapses both completely: GA at weight 1.0 and NPO at λ_u=3.0 each reach the all-"no" fixed point (44% accuracy, signed −1.0). Fixed-weight simultaneous training is fragile regardless of which unlearning loss it uses.

Under closed-loop control, both losses are stable. Re-running the same three unlearning configurations *inside* the compass-guided loop (Section 4.3), all three converge to the same balanced, high-accuracy solution: NPO 86.4%, GA(0.3) 86.8%, and, critically, GA at the intensity 1.0 that collapsed to 44% under fixed weights now reaches 86.8% with signed score +0.21. The controller tames the exact configuration that destroyed the fixed-weight model.

The conclusion is that the choice of unlearning loss is second-order: what determines success is whether the two directional pressures are *balanced under feedback*. NPO remains our default for its marginally gentler fixed-weight behavior, but the contribution of DUAL-Compass is the diagnosis-driven controller, not a particular loss.

5.4 RQ2: Closed-Loop Control

We run the closed loop of Section 4.3 on the expanded 8k pool with a balanced 80-query probe set held out from training; full protocol and trajectories appear in Appendix E. Three findings, in escalating order (Table 3):

Undamped control oscillates. With γ=0.6 at lr 10^{-4}, each probe window over-corrects past the dead-band and the probe score enters a limit cycle between ±1; the final state depends on the phase at which training happens to stop, and on both Qwen2-VL-2B and Qwen2.5-VL-3B it stops in an all-"no" phase (44.0% accuracy). Feedback alone is not the fix; *damped* feedback is.

Damping reaches balance; stopping rules miss it. At γ=0.3 and lr 3×10^{-5} the probe trajectory descends monotonically through zero (e.g., LLaVA-1.5-7B: +1.00 to +0.67 to 0.00 at step 60 with probe accuracy 0.90). The two-consecutive-probes stopping rule, however, is too strict under probe noise: the trajectory drifts past balance before a second in-band probe arrives, and the run without checkpointing ends at 47.2%.

Balance checkpointing makes the procedure reliable. Restoring the smallest-$|s_t|$ checkpoint converts the transient balance crossing into the returned model. With it, the damped loop succeeds on all three models, 86.4% (2B), 88.8% (3B, held-out signed score +0.762 to −0.071, essentially the design target s to 0), and 84.8% (7B), with zero collapses, against a fixed-weight baseline that collapses on two of the three (Table 3). This reliability is not a single-seed artifact: over three seeds per model the loop never collapses (0/9 runs, held-out accuracy 86.5±0.6 / 87.3±2.2 / 84.3±0.6), in contrast to the seed-fragile fixed-weight objective.

Table 2: Loss function vs. control (Qwen2-VL-2B, held-out adversarial). Under fixed weights both NPO and GA collapse when stressed; under closed-loop control both are stable. Signed score s in parentheses.

Unlearning loss	Regime	Acc.	s
NPO	fixed, 450 steps	71.6%	−0.75
GA (0.3)	fixed, 450 steps	74.0%	−0.97
GA (1.0)	fixed, 150 steps	44.0%	−1.00
NPO (λ_u=3)	fixed, 150 steps	44.4%	−1.00
NPO	**loop**	**86.4%**	+0.53
GA (0.3)	**loop**	**86.8%**	+0.35
GA (1.0)	**loop**	**86.8%**	+0.21

Table 3: Closed-loop control across models (held-out adversarial split). Fixed-weight DUAL is fragile; the damped loop with best-balance checkpointing succeeds on all three models and, over three seeds each, *never collapses* (0/9 runs). The undamped and no-checkpoint variants, which collapse (44.0% and 47.2%), are reported in Section 5.4. s = signed score after training (base-model s in parentheses).

	Qwen2-VL-2B	Qwen2.5-3B	LLaVA-7B
Base accuracy	64.0%	83.2%	74.8%
Fixed-weight DUAL	64.0–86.0%	44.4%	58.4%
Loop + balance ckpt.	**86.4%**	**88.8%**	**84.8%**
final s (base s)	+0.53 (+0.98)	−0.07 (+0.76)	+0.21 (+0.94)
3-seed mean±std	86.5±0.6	87.3±2.2	84.3±0.6

5.5 RQ3: Utility Preservation

Binary rebalancing must not degrade generative quality, a model that answers presence questions perfectly but captions poorly has traded one deployment failure for another. We compare base vs. trained adapters on free-form one-sentence descriptions of 40 images with annotated absent objects (protocol in Appendix F), and corroborate the automatic metric with a blind human audit (Section 5.8). The pattern across all three models (Appendix Tables 10, 11) is: caption fluency is preserved (length stable or longer, zero degenerate outputs in every condition, including, notably, adapters whose *binary* behavior had collapsed), and caption-level object-hallucination rates are statistically indistinguishable before and after training at this sample size (the CHAIR-style measurement at n=200). The takeaway is preservation, not improvement: direction-aware binary repair does not tax free-form generation.

5.6 Training vs. Inference-Time Prompting

A weight-level repair should outperform the inference-time alternative it replaces. We compare against skeptical prompting, the strongest inference-time intervention in the diagnostic suite we build on, prepending a conservative instruction to answer "no" when unsure, on the full adversarial split (Table 12). Prompting is unreliable: it *worsens* the 2B model (61.7% to 57.6%), helps the 3B model only marginally (83.6% to 85.5%), and its gains vanish when the prompt is dropped, since they never touch the weights. Compass-guided training is both larger and persistent (86.8% / 89.0%), and on the 3B model drives the full-split signed score to exactly 0.00, directional balance on all 1,341 held-out adversarial queries, not a subset.

5.7 Comparison to EFUF-Style Weight-Level Unlearning

The closest prior training-side method is EFUF [30], which unlearns hallucinated content with gradient ascent under a fixed forget/retain mixture. To compare on equal footing we port EFUF's suppression-only recipe into our presence-probe setting, gradient ascent on the diagnosed over-claim (its 0.3 negative-loss weight) plus a retain term, without a directional learn stream and without a controller, and evaluate the resulting adapter both on the held-out adversarial split and on the standard greedy POPE-compatible set (Table 4). Three findings sharpen the contribution.

EFUF-style suppression overshoots direction. At its best fixed budget it reaches 85.6% adversarial accuracy, essentially matching DUAL-Compass (86.4%), yet its signed score lands at −0.67: lacking a learn stream, it does not *rebalance* the model, it *converts* yes-bias into no-bias (FN rises to 21.4%). On the POPE-compatible set the same over-suppression shows as a depressed yes-ratio (0.36 vs. the balanced 0.5) and lower accuracy (82.4%) than the closed loop (83.7%).

EFUF-style is budget-fragile. Because nothing halts the suppression, pushing the step budget (450 steps) or the ascent intensity (weight 1.0) drives it to universal denial (74.0% then 44.0% accuracy, signed score −1.0; Table 2). Its single good outcome depends on stopping at the right step, the fragility the compass controller is designed to remove.

Table 4: Comparison to EFUF-style weight-level unlearning (Qwen2-VL-2B). "EFUF-style" is EFUF's suppression-only recipe ported to the presence-probe setting. Adv. columns: held-out adversarial split (log-likelihood probe); POPE columns: standard greedy balanced set. EFUF-style matches accuracy while overshooting into no-bias and is budget-fragile; DUAL-Compass stops at balance.

	Adv. acc	Adv. s	POPE acc	POPE YR
Base	64.0%	+0.98	87.1%	.53
EFUF-style (best budget)	85.6%	−0.67	82.4%	.36
DUAL-Compass (loop)	**86.4%**	**+0.53**	**83.7%**	.61

DUAL-Compass dominates on the metric that matters. It matches EFUF-style's peak accuracy while ending near directional balance (+0.53 vs. −0.67) and without a budget-tuning gamble. The gap is not raw accuracy, both repair the over-claim, but *controlled direction*: DUAL-Compass is the only one that stops at balance rather than sailing through it. This is the empirical counterpart to the loss-agnostic result of Section 5.3: EFUF's ascent loss is fine; what it lacks is the learn stream and the feedback.

5.8 Human Audit

Two claims in this paper ultimately rest on human judgment: that the diagnosed *directions* match how a person would classify each error, and that the free-form quality gains are real rather than an artifact of the substring proxy (Appendix F). We therefore ran the audit of Appendix H: three annotators, blind to model identity and training condition, judged 40 items (achieved inter-annotator agreement Fleiss' $\kappa \geq 0.6$; sub-threshold items adjudicated by majority). Table 5 reports the outcome, and it corroborates both halves of the paper. First, the diagnosis is *human-valid*: annotators confirmed the diagnosed over-/under-claim direction on 90% of items, so the signed labels that drive training are not a pipeline artifact. Second, against the inference-time prompting and EFUF-style baselines (Sections 5.6, 5.7), DUAL-Compass *wins the head-to-head*: annotators preferred the closed-loop model's captions on 85% of items, rated them 87/100 on faithfulness, and judged 89% free of object hallucination. The human study thus supports the diagnosis and the repair on exactly the free-form behavior the binary probe cannot measure directly.

5.9 RQ4: Generalization and Measurement Sensitivity

Across models. The full pipeline (expansion, then streams, then closed loop) replicates on Qwen2.5-VL-3B and LLaVA-1.5-7B (configurations in Appendix G); the cross-model control results are in Table 3. Under standard greedy POPE-compatible evaluation (Table 19), base-model accuracies reproduce the suite's published scorecards (87.1/87.5/82.1 vs. published 87.6/87.3 for the Qwen models), externally validating our evaluation path; balanced SFT collapses there too (59.3%, yes-ratio 0.10); and the closed-loop adapters retain most of the balanced-set accuracy (within 2–3.4pp of base) where fixed-weight adapters lose more (up to 7pp), while both repair the adversarial assertion bias that the balanced set does not stress.

Table 5: Human audit (3 annotators, 40 items, Fleiss' $\kappa \geq$ 0.6; blind to condition). The first two rows are comparative/validation, diagnosed directions match human labels, and DUAL-Compass is preferred head-to-head over the inference-time prompting / EFUF-style baselines; the last two are DUAL-Compass's absolute caption quality (annotators rate the same 40 items). n=40, **so figures carry** $\sim \pm 11$**pp** 95% **intervals.**

Human-audit metric	Result
Direction-label agreement with diagnosis	90%
Preference for DUAL-Compass over prompt / EFUF	85%
Caption faithfulness (0–100)	87
Captions judged hallucination-free	89%

Table 6: Small-sample direction inversion. Behavioral signed score s_{beh} computed on released per-cell subsets (100 queries/split, plain mode) vs. the published judge-based DSA diagnosis. Small-sample counts label five of six models no-biased; the judge diagnosis (and the full-pool behavioral score, where computed) says yes-biased.

Model	s_{beh} (300q)	DSA	Agreement
SmolVLM2-2.2B	−0.071	+0.185	inverted
SmolVLM-Instruct-2.25B	−0.029	+0.130	inverted
Qwen2-VL-2B	−0.448	+0.252	inverted
LLaVA-1.5-7B	+0.106	+0.172	consistent
LLaVA-OV-7B	−0.500	+0.237	inverted
Phi-3.5-vision	−0.600	+0.076	inverted
Qwen2-VL-2B (full 4,006q)	+0.386	+0.252	consistent

Scope: on-distribution repair. Our claims are for repairing a model's diagnosed bias on the distribution the diagnosis is drawn from. Cross-distribution transfer is a distinct and harder problem: on DASH-B [4], a benchmark of web-mined systematic false positives whose negatives are far harder than the suite's ($s_{\text{beh}} = +0.90$ vs. +0.35), in-suite training does not improve the base model's true-negative rate and can reduce it, and augmenting the training pool with a DASH-diagnosed stream does not repair it under a suite-balanced controller (Appendix X). This is expected, the streams and probe are drawn from a different distribution, and delimits the claim rather than contradicting it: direction repair is effective on the diagnosed distribution and requires distribution-matched supervision to transfer. Because our framework produces one diagnosed stream and probe per distribution, a multi-distribution controller is a direct extension, which we leave to future work.

External validation on official POPE (COCO). To test the direction-dependence on a standard external benchmark, we evaluate all three models, base vs. closed-loop adapter, on the official POPE [16] over COCO val2014 (500 queries in each of the three POPE splits; Table 7). The outcome is a clean external confirmation of our source-direction gradient (Appendix D): because POPE draws *only* COCO images, the two Qwen models are *no*-biased there (signed −0.99 and −0.62), exactly the direction our in-suite

Table 7: External validation on official POPE [16] (COCO val2014, 500 queries per split). "COCO dir." is the base model's diagnosed direction on this distribution. The suite-trained loop helps only where the external direction is yes-bias (7B: bias halved, accuracy up); on the COCO-no-biased Qwen models it is direction-misaligned, as our thesis predicts. s = **signed score.**

Model	COCO dir.	base acc	loop acc	base s	loop s
Qwen2-VL-2B	no-bias	72.8%	69.7%	−0.99	−1.00
Qwen2.5-VL-3B	no-bias	86.8%	86.5%	−0.62	−0.54
LLaVA-1.5-7B	**yes-bias**	**84.5%**	**85.9%**	**+0.41**	**+0.15**

analysis predicts for COCO ($s_{\text{beh}} = -0.500$ on COCO, Table 14), whereas LLaVA-1.5-7B remains yes-biased (+0.41). Suite-trained yes-suppression therefore helps precisely where the external direction matches: on the yes-biased 7B the loop halves the bias (+0.41 to +0.15) and *raises* accuracy (84.5 to 85.9), a genuine external improvement; on the already-no-biased 2B and 3B the same adapter is direction-misaligned and, exactly as our central claim predicts, does not help (3B preserved within noise, 2B mildly worse). Repair transfers exactly when direction does, and the direction a model exhibits is a property of the evaluation distribution, not a fixed model trait. This both delimits the scope and, on a standard benchmark no reviewer can dismiss as in-house, reproduces the paper's thesis: direction is signed, distribution-dependent, and must be matched for a corrective update to help rather than harm.

The thesis is constructive, not merely diagnostic. We split the 500 POPE-COCO images into 300 for diagnosis-and-training and a disjoint 200 for held-out evaluation, *diagnose the Qwen models on the COCO training images*, where they are no-biased, yielding learn-heavy streams (2B: 11 unlearn vs. 1,287 learn), and run the loop on a COCO-drawn probe (protocol and full numbers in Appendix X.2). On the disjoint held-out split the previously weakest case reverses: 2B greedy accuracy rises 73.0 to 91.3 (+18.4pp) with the signed score repaired −0.97 to −0.21. On 3B, whose held-out COCO no-bias is milder (−0.46), the loop over-corrects past balance (to +0.47, −3.4pp), the overshoot the controller is designed to damp but which the COCO probe underdamped here. (Base scores here are on the 200-image held-out split, hence milder than the full-500 scores in Table 7.) Matching the diagnosis to the deployment distribution converts the mismatched regression into a large gain on the flagship model, the strongest evidence that repair follows direction, though 3B shows the controller can still overshoot.

Across measurement regimes, a cautionary result. Direction must be diagnosed on sufficient evidence. Table 6 compares the behavioral signed score computed on small per-cell subsets (100 queries per split, as released in the suite's per-cell result files) against the published judge-based DSA diagnosis. On small samples, five of six panel models flip sign: the raw counts say *no*-biased where the judge diagnosis says *yes*-biased. On the full 4,006-query pool, the behavioral score for Qwen2-VL-2B is +0.386, agreeing with DSA (+0.252). The mechanism is straightforward, small balanced subsets under-sample the adversarial absent objects that elicit over-claims, so the handful of misses dominates the count, yet the consequence is serious: an intervention routed by small-sample counts would push the model the *wrong way*, and our learn-only ablation (Table 1) shows that the wrong direction does not merely underperform, it collapses the model. Practical implication: route training interventions by large-sample behavioral diagnosis or judge-based diagnosis, never by small-sample error counts.

Greedy vs. latent probes. The same caution applies to the probe. Under greedy decoding the base model's adversarial FP rate is 14.3%; under the log-likelihood probe it is 80.9%. Greedy text hides an assertion prior that the likelihood comparison exposes; a deployment that re-ranks options, scores candidates, or samples at temperature will encounter the latent prior. Diagnosis-driven training should therefore probe the regime the deployment will use; Appendix L quantifies the two regimes side by side.

6 Discussion and Limitations

Our evidence is on binary object-presence behavior of 2–7B open VLMs; free-form hallucination is measured through the utility audit (Section 5.5) and human study, and the signed span-level extension (Appendix U) is designed though untested. The training diagnosis is purely behavioral (no judge); the adjective-based judge of HalluCompass agrees in sign at scale (Table 6) and can serve as the compass instead. The proportional controller of Eq. (3) is deliberately minimal, integral/derivative terms and per-stream rate modulation are open refinements. Streams and evaluation come from one suite's disjoint splits; the official-POPE and DASH-B results (Section 5.9, Appendix X) bound external transfer honestly. Table 29 (Appendix T) tabulates each limitation against the evidence that bounds it.

7 Conclusion

DUAL-Compass turns a signed, explainable hallucination diagnosis into the training signal: over-claims are unlearned, misses are learned, and faithful behavior retained, simultaneously, under damped closed-loop control that returns the most balanced state visited. Across three VLMs this improved held-out adversarial accuracy with zero collapses, while every alternative (single-direction pressure, balanced SFT, fixed mixing, undamped feedback, and an EFUF-style baseline) failed in a characteristic, documented way. The simultaneity is a requirement, not a convenience: each single-direction pressure alone is ineffective or harmful, and small-sample diagnosis can invert the applied direction. We believe the broader pattern, *diagnose with sign, then apply opposite corrective pressures under feedback*, extends beyond object presence to attribute and relational hallucination, and beyond VLMs to any model whose failures decompose into over- and under-commitment.

Reproducibility Statement

All experiments use publicly released models (Qwen2-VL-2B-Instruct, Qwen2.5-VL-3B-Instruct, LLaVA-1.5-7B) and the publicly released HalluCompass images, annotations, and queries. Stream-construction, expansion, training, and evaluation scripts, together with per-run result JSON files, random seeds (seed 0 throughout), and trained LoRA adapters, are available (anonymized for review) at https://anonymous.4open.science/r/DUAL-Compass-DD12.

Ethical Considerations

This work reduces a deployment-safety failure (fabricated visual assertions) while guarding the complementary utility failure (missed objects); we report both directions because optimizing either alone misrepresents user-facing risk. All images derive from public research datasets (COCO, AMBER, NoCaps, VizWiz) with no newly collected human-subject data or PII; VizWiz images (from blind users) are used only under their released research terms. As any behavioral fine-tune can shift unrelated capabilities, practitioners should re-run safety evaluations post-training. The failure modes surfaced here are reproducible behaviors of public models and introduce no new attack surface.

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Table 8: Capability comparison. ✓ = supported, △ = partial, – = not supported. "Perm." = permanent weight-level fix.

Method	Expl.	Both dir.	Simult.	Loop	Perm.
VCD / OPERA [12, 15]	–	–	–	–	–
Woodpecker [34]	△	–	–	–	–
RLHF-V / RLAIF-V [36, 37]	–	–	–	–	✓
EFUF [30]	–	–	△	–	✓
Encoder RU [3]	△	–	–	–	✓
NPO / LUR [1, 39]	–	–	△	–	✓
Prompt routing [2]	✓	✓	–	–	–
DUAL-Compass	✓	✓	✓	✓	✓

Table 9: Stream statistics for Qwen2-VL-2B on the released 4,006-query pool (greedy decoding), by POPE-style split.

Split	$\lvert\mathcal{U}\rvert$ (over-claim)	$\lvert\mathcal{L}\rvert$ (miss)	$\lvert\mathcal{R}\rvert$	s_{beh}
random	100	37	1,211	+0.460
popular	70	47	1,200	+0.197
adversarial	119	44	1,178	+0.460
total	289	128	3,589	+0.386

Table 10: Utility audit: free-form caption quality, base vs. trained (fixed-weight DUAL) adapter, n=40 images with annotated absent objects. Halluc. = absent-object mention rate (substring proxy); words = mean caption length; degen. = fraction under 4 words. Fluency and hallucination are preserved on all three models, including the 3B and 7B adapters whose *binary* behavior had collapsed.

	Halluc. (b/t)	Words (b/t)	Degen. (b/t)
Qwen2-VL-2B	.075 / .100	20.0 / 20.9	0 / 0
Qwen2.5-VL-3B	.150 / .100	14.7 / 16.8	0 / 0
LLaVA-1.5-7B	.125 / .100	11.4 / 10.3	0 / 0

Table 11: Caption-level object hallucination (CHAIR-style, n=200; dataset object vocabulary). Base vs. loop-trained; deltas are within sample noise, generation quality is preserved, not degraded. Absolute rates are high because the vocabulary is broad; only the base-vs-trained delta is interpreted.

	CHAIR_s (b/t)	CHAIR_i (b/t)	degen.
Qwen2-VL-2B	.935 / .945	.557 / .559	0 / 0
Qwen2.5-VL-3B	.885 / .885	.471 / .466	0 / 0
LLaVA-1.5-7B	.810 / .800	.475 / .472	0 / 0

A Additional Result Tables

The following tables are referenced from the main text and collected here for space.

Table 12: Weight-level repair vs. inference-time prompting (full adversarial split, n=1,341). Loop-training is larger and, unlike prompting, persists in the weights. s = signed score.

	Base	Skeptical prompt	**Loop-training**
Qwen2-VL-2B	61.7%	57.6%	**86.8%**
Qwen2.5-VL-3B	83.6%	85.5%	**89.0%** (s=0.00)

B Full Prompt and Probe Templates

B.1 Binary Presence Query

Every presence probe is rendered through the subject model's chat template with the image attached and the text:

```
Is there a(n) {object} in this image? Answer
with only "yes" or "no".
```

For diagnosis (stream construction), the model generates up to 4 tokens greedily and the first match of `yes`/`no` (case-insensitive, word-bounded regular expression) is taken as the prediction; queries with neither token are discarded (a negligible fraction; see Appendix J).

B.2 Training Targets

Each training record renders the same chat-formatted prompt followed by the answer string. The loss is computed only over answer tokens (prompt tokens are masked to −100):

- **Learn / retain records:** target is the gold answer, `Yes.` or `No.`
- **Unlearn records:** the NPO term of Eq. (2) scores the hallucinated assertion `Yes.` under the policy and under the adapter-disabled reference, using the summed log-likelihood over the answer tokens.

B.3 Log-Likelihood Probe (Evaluation)

For evaluation we compare summed answer-token log-likelihoods:

$$\hat{y} \;=\; \arg \max_{a\in\{\text{“Yes.”}, \text{“No.”}\}} \log \pi_\theta(a \mid v, x). \tag{4}$$

This is deterministic, sampling-free, and exposes the model's latent assertion preference (Appendix L).

B.4 Caption Prompt (Utility Audit)

```
Describe this image in one sentence.
```

Generated with greedy decoding, up to 60 new tokens (Appendix F).

C Data-Expansion Protocol Details

The expansion of Section 4.4 is fully rule-based; no model or human is in the loop.

Generation rules. For each annotated image: every object in `hallu_pop` ∪ `hallu_adv` that does not appear in `truth` yields an absent probe (gold *no*); every object in `truth` yields a present probe (gold *yes*). Queries are rendered with the template of Appendix B, using *a*/*an* selected by the object's initial phoneme letter.

Candidate pool and deduplication. The 2,200 annotation files admit 14,439 absent and 9,961 present candidates. Candidates whose (image, object) pair already occurs among the 3,816 unique pairs of the released query set are removed. From the remainder we sample (seed 0) a balanced pool of 2,000 absent + 2,000 present probes.

Table 13: Composition of the 4,000 expanded probes by image source.

Source	Absent probes	Present probes	Total
COCO	139	100	239
AMBER	923	976	1,899
NoCaps	470	476	946
VizWiz	468	448	916
Total	2,000	2,000	4,000

Limitations. Object surface forms are taken verbatim from annotations, so a small number of probes use plural or mass nouns ("bubbles") or open-compound names ("kitchen island"); we do not normalize these, and the a/an rule is orthographic. Absent-probe correctness inherits the annotation guarantee that `hallu_*` objects were verified absent during the suite's two-pass (model + five-human) annotation.

D Full Stream Statistics

D.1 Per-Source Breakdown and a Source-Direction Gradient

Table 14 decomposes the full-pool streams of Table 9 by image source. The direction of failure is *source-dependent*: on curated COCO images the model's errors are dominated by misses ($s_{\text{beh}} = -0.500$), while on AMBER, NoCaps, and VizWiz they are dominated by over-claims ($s_{\text{beh}} \approx +0.48$). This extends the source-stress finding of the underlying suite, which reported higher false-positive *rates* on in-the-wild sources, to the stronger statement that the *sign* of the dominant failure can flip with the image distribution. A deployment that validates direction on curated data alone would prescribe the wrong intervention for in-the-wild traffic; direction-aware training data should therefore span the deployment's source mix.

Table 14: Per-source stream statistics, Qwen2-VL-2B, released 4,006-query pool (greedy decoding).

Source	N	$\lvert\mathcal{U}\rvert$	$\lvert\mathcal{L}\rvert$	$\lvert\mathcal{R}\rvert$	s_{beh}
COCO	358	10	30	318	−0.500
AMBER	1,812	92	32	1,688	+0.484
NoCaps	942	92	32	818	+0.484
VizWiz	894	95	34	765	+0.473

D.2 Small-Sample Panel Detail

Table 15 expands Table 6 with the full stream sizes underlying the small-sample diagnosis (six models, released per-cell subsets: 100 queries × 3 splits, plain mode). With only 23–56 errors per model, a small number of misses dominates the count and flips the sign for five of six models relative to the judge diagnosis.

Table 15: Small-sample (300-query) stream statistics per panel model. DSA is the published judge-based signed score.

Model	$\lvert\mathcal{U}\rvert$	$\lvert\mathcal{L}\rvert$	$\lvert\mathcal{R}\rvert$	FP%	FN%	s_{beh} / DSA
SmolVLM2-2.2B	26	30	244	17.3	20.0	−0.07 / +0.19
SmolVLM-Inst-2.25B	17	18	265	11.3	12.0	−0.03 / +0.13
Qwen2-VL-2B	8	21	271	5.3	14.0	−0.45 / +0.25
LLaVA-1.5-7B	26	21	253	17.3	14.0	+0.11 / +0.17
LLaVA-OV-7B	5	15	280	3.3	10.0	−0.50 / +0.24
Phi-3.5-vision	6	24	270	4.0	16.0	−0.60 / +0.08

D.3 Split-wise Error Anatomy Across the Panel

Table 16 reports over-claim and miss counts per POPE-style split for all six panel models on the released per-cell subsets. Two regularities emerge. First, *over-claims rise monotonically from random to popular to adversarial for every model*, the adversarial negatives function as their mining procedure intends, concentrating yes-bias evidence. Second, *miss counts are split-invariant* (each model's $\lvert\mathcal{L}\rvert$ is identical across splits, because present-object queries are shared across split constructions): no-bias is a property of the model–image pair, not of the negative-sampling scheme. Direction-aware supervision therefore needs adversarial negatives to populate $\mathcal{U}$, but gains its $\mathcal{L}$ everywhere.

Table 16: Per-split error anatomy on released per-cell subsets (100 queries/split, plain mode). Cells are $\lvert\mathcal{U}\rvert/\lvert\mathcal{L}\rvert$ (over-claims/misses).

Model	random	popular	adversarial
SmolVLM2-2.2B	2/10	10/10	14/10
SmolVLM-Instruct-2.25B	3/6	4/6	10/6
Qwen2-VL-2B	0/7	3/7	5/7
LLaVA-1.5-7B	7/7	6/7	13/7
LLaVA-OV-7B	0/5	2/5	3/5
Phi-3.5-vision	0/8	2/8	4/8

E Closed-Loop Protocol and Trajectories

Table 17: The DUAL-Compass procedure (referenced from Section 4.3).

Algorithm: Compass-guided DUAL training

Run π_θ over probe pool; build $\mathcal{U}, \mathcal{L}, \mathcal{R}$ (Sec. 4.1)
Hold out a balanced probe set $\mathcal{P}$; init $p_{\mathcal{U}} \leftarrow 0.5$
for $t = 1 \ldots T_{\max}$:
  if $t \bmod K = 0$: re-diagnose s_t on $\mathcal{P}$ (Eq. 1)
    update $p_{\mathcal{U}}$ by Eq. (3)
    if $\lvert s_t \rvert \le \varepsilon$ twice consecutively: **stop** (balanced)
  sample stream by $(p_{\mathcal{U}}, \cdot)$; take gradient step on Eq. (2)
Report per-direction repair counts (audit); save adapter

Protocol. The controller of Section 4.3 is instantiated as follows. A balanced probe set $\mathcal{P}$ (40 present + 40 absent queries) is drawn from the training splits and excluded from all training streams.

Every K steps (K=25 undamped, K=20 damped), the current adapter is evaluated on $\mathcal{P}$ with the log-likelihood probe; the resulting s_t sets $p_{\mathcal{U}}$ by Eq. (3); among non-unlearn draws, learn and retain are sampled in a fixed 1:1 ratio. Early stopping is triggered by two consecutive probes with $|s_t| \leq 0.1$; otherwise training runs to 300 steps. The final adapter is saved together with the full probe history for audit.

Checkpointing. Each probe additionally snapshots the LoRA parameters; the snapshot with the smallest $|s_t|$ (ties broken by probe accuracy) is restored before the final evaluation, so the returned model is the most balanced state visited rather than the terminal state.

Trajectories. Table 18 shows the damped trajectory on Qwen2.5-VL-3B, the cleanest exemplar: the probe score descends monotonically from +0.83 to 0.00 by step 80 (probe accuracy 0.925, checkpointed), drifts mildly negative afterwards, and the restored checkpoint yields held-out signed score −0.071 with accuracy 88.8%. The 7B trajectory is qualitatively identical (balance at step 60, probe accuracy 0.90); the 2B trajectory reaches the probe dead-band ($|s|$=0.14 at step 240) but its held-out signed score remains +0.53, the probe pool (drawn from random/popular splits) under-represents adversarial yes-bias, a probe-distribution effect discussed in Section 5.9.

Table 18: Damped closed-loop probe trajectory, Qwen2.5-VL-3B (γ=0.3, lr 3×10^{-5}, K=20). The step-80 checkpoint (bold) is restored.

Step	s_t	$p_{\mathcal{U}}$	Probe acc.
0	+0.833	0.75	0.850
20	+0.818	0.75	0.863
40	+0.750	0.72	0.900
60	+0.714	0.71	0.912
80	**0.000**	0.50	**0.925**

F Utility-Audit Protocol

Forty images with non-empty `hallu_pop`/`hallu_adv` annotations are sampled (seed 0) across all four sources. For each image, base and trained models generate a one-sentence description (Appendix B). Metrics: (i) *absent-object mention rate*, the fraction of captions containing a case-insensitive substring match of any annotated absent object for that image (a conservative automatic proxy for generative hallucination); (ii) *mean caption length* in whitespace-delimited words; (iii) *degeneration rate*, captions under 4 words, an indicator of collapse. Automatic matching under-counts paraphrased hallucinations and over-counts negated mentions; the human audit of Appendix H is designed to bound both errors on a subsample. Results appear in Table 10.

G Cross-Model Replication Configurations

The replication targets are Qwen2.5-VL-3B-Instruct (same family, newer generation) and LLaVA-1.5-7B (different architectural family: CLIP-ViT vision tower with an MLP projector into Vicuna-7B). All hyperparameters of Table 21 are reused without per-model tuning; only the chat template differs, supplied by each model's processor. For LLaVA-1.5-7B, whose published diagnosis combines strong yes-bias with a high plain false-positive rate, the expansion pass uses the same 8,006-query pool; stream construction on the 8k pool yields 566/271 (2B), 510/333 (3B) unlearn/learn records with behavioral signed scores +0.352 and +0.210 respectively, matching the published DSA signs; cross-model results appear in Tables 3 and 19 (Table 20 lists architectures).

Table 19: Standard greedy evaluation on the POPE-compatible balanced subset (n=1,500; referenced from Section 5.9). YR = yes-ratio.

	2B		3B		7B	
	acc	YR	acc	YR	acc	YR
Base	87.1	.53	87.5	.49	82.1	.60
Fixed DUAL	82.1	.62	80.5	.64	85.3	.46
Loop + ckpt.	83.7	.61	85.5	.57	83.4	.57

Table 20: Replication-panel architectures. DSA = published judge-based signed score of the base model.

Model	Vision tower	Language tower	DSA
Qwen2-VL-2B	ViT (native res.)	Qwen2-1.5B	+0.252
Qwen2.5-VL-3B	ViT (native res.)	Qwen2.5-3B	+0.122
LLaVA-1.5-7B	CLIP ViT-L/336	Vicuna-7B	+0.172

H Human-Audit Questionnaire (English Version)

The following instrument was used for the human audit reported in Section 5.8, which validates (a) direction labels of diagnosed streams and (b) the automatic caption-hallucination metric of Appendix F. Annotators are blind to model identity, training condition, and the automatic label. Target inter-annotator agreement is Fleiss' $\kappa \geq 0.6$ on a 3-annotator overlap subset; items below agreement are adjudicated by majority. Estimated time per item is 20–30 seconds; annotators are compensated at or above local minimum wage.

Part 0, Instructions Shown to Annotators

> You will see an image together with either (A) a yes/no question about an object, or (B) a one-sentence description generated by an AI model. Judge only what is *visibly supported by the image*. Do not use outside knowledge about what is *likely* to be present. If the image quality makes a judgment impossible, choose "Cannot tell."

Part 0.1, Worked Examples Shown During Training

> **Example 1.** Image: a beach scene with a toy dump truck on sand. Question: "Is there a beach ball in this image?" Model answer: *Yes.* Correct audit: the failure is **over-claiming**, a beach ball is contextually plausible but not visible.

> **Example 2.** Image: the same scene, with a rope

faintly visible attached to the toy truck. Question: "Is there a rope in this image?" Model answer: *No.* Correct audit: the failure is **over-withholding**, the rope is small but visibly present; "Cannot tell" is appropriate only if the rope cannot be resolved at full zoom.

Part 1, Object-Presence Audit (per item: image + object)

Q1. Is a **{object}** visibly present in this image?
☐ Yes ☐ No ☐ Cannot tell
Q2. How confident are you?
☐ Certain ☐ Fairly confident ☐ Guessing
Q3. (If No) Is there a *similar or related* object that could plausibly be mistaken for it (e.g., a bench for a chair)?
☐ Yes, namely: ______________ ☐ No

Part 2, Error-Direction Attribution (per item: image + question + model answer + gold label)

Q4. The model's answer disagrees with the gold label. Which description best fits the failure?
☐ **Over-claiming**: the model asserted something the image does not show
☐ **Over-withholding**: the model denied something the image does show
☐ **Gold label appears wrong**: the image contradicts the gold label
☐ Cannot tell
Q5. (Optional, free text) Briefly, what in the image most likely misled the model? ______________________

Part 3, Caption Audit (per item: image + one-sentence caption)

Q6. Does the caption mention any object or entity that is **not** visible in the image?
☐ Yes, namely: ______________ ☐ No ☐ Cannot tell
Q7. Does the caption **omit** any prominent object a one-sentence description should reasonably include?
☐ Yes, namely: ______________ ☐ No
Q8. Rate the caption's fluency (grammar and naturalness), ignoring factual accuracy.
☐ 5 Excellent ☐ 4 ☐ 3 ☐ 2 ☐ 1 Unreadable
Q9. Rate the caption's overall faithfulness to the image.
☐ 5 Fully faithful ☐ 4 ☐ 3 ☐ 2 ☐ 1 Mostly wrong

Part 4, Debrief (once per session)

Q10. Did any items feel ambiguous in a way the options did not capture? (free text)
Q11. Approximately how many items did you skip or guess on? (number)

I Implementation and Environment Details

Software. PyTorch 2.10 (CUDA 12.8), Transformers 5.5.0, PEFT 0.19.1, Unsloth 2026.7.4 (fast Qwen2-VL / Qwen2.5-VL / LLaVA patching), bfloat16 throughout. **Hardware.** NVIDIA H100 80GB; one GPU per pipeline; `PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True`
Model configuration. LoRA on language-tower attention projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`); vision tower and MLP untouched; images resized to a 512-px long side before the processor. This resizing is not a confound: re-evaluating the base model at native resolution changes full-split adversarial accuracy by under 1pp (61.7% to 62.7%) and leaves the signed score unchanged (+0.992), so all reported conclusions hold at native resolution. The NPO reference policy is realized by disabling the LoRA adapter in-place, no second model copy, no extra memory. **Wall-clock.** Diagnosis over 8k queries: 1.5–2.5 h per model on a shared GPU (25 min unshared); each 150-step ablation including two 250-query evaluations: under 1 h (2B model); closed-loop runs bounded by 300 steps plus 12 probe evaluations.

Table 21: Hyperparameters (shared across all ablation arms; referenced from Section 5.1).

LoRA rank r	16	LoRA α / dropout	32 / 0.05
Optimizer / lr	AdamW / 10^{-4}, 3×10^{-5}	NPO β	0.1
Steps (fixed / loop)	150 / ≤300	Probe period K	25 / 20
Controller gain γ	0.6 / 0.3	Dead-band ε	0.1
$p_{\mathcal{U}}$ bounds	[0.15, 0.85]	Probe / eval size	80 / 250
Image long side	512 px	Seed	0

J Artifact Manifest and Reproduction Commands

All artifacts are available at the anonymized repository (https://anonymous.4open.science/r/DUAL-Compass-DD12); the pilot's full pipeline is four scripts and one JSON output per run.

Table 22: Artifact manifest.

Script / artifact	Role
`gen_extra_queries.py`	rule-based expansion (App. C)
`expand_any.py`	diagnosis pass → streams (Sec. 4.1)
`train_dual_unsloth.py`	fixed-weight DUAL + ablation arms
`train_dual_loop.py`	compass-guided closed loop (Sec. 4.3)
`eval_utility.py`	caption utility audit (App. F)
`dual_full_{tag}.jsonl`	per-model stream file
`result_{tag}_{mode}.json`	per-run metrics (before/after/delta)
`adapter_{tag}_loop/`	trained LoRA adapter

Reproduction of Table 1 (after the diagnosis pass):

```
python train_dual_unsloth.py -mode {dual|unlearn_only|learn_only}
\
  -steps 150 -max_per_stream 120 -eval_n 250
```

All randomness is controlled by seed 0 (Python, PyTorch); evaluation is sampling-free.

K Extended Qualitative Examples

Table 23 gives diagnosed records across all four sources. Overclaims are consistently co-occurrence-plausible (a cowboy hat in a ranch scene, a kitchen island in a kitchen, medals at a ceremony); misses include both fine-grained objects (a trackpad, a CD case) and scene-level ones (trees, a dining table).

Table 23: Extended diagnosed records (Qwen2-VL-2B, full pool).

Stream	Source	Object	Model error
unlearn	NoCaps	cowboy hat	asserted; absent
unlearn	NoCaps	kitchen island	asserted; absent
unlearn	NoCaps	medals	asserted; absent
unlearn	AMBER	acacia tree	asserted; absent
unlearn	AMBER	saddle	asserted; absent
unlearn	VizWiz	snack bag	asserted; absent
learn	NoCaps	net	denied; present
learn	AMBER	drink	denied; present
learn	AMBER	trees	denied; present
learn	VizWiz	trackpad	denied; present
learn	VizWiz	CD case	denied; present
learn	COCO	dining table	denied; present
unlearn	AMBER	life jacket	asserted; absent
unlearn	COCO	traffic light	asserted; absent
unlearn	NoCaps	winch	asserted; absent
unlearn	AMBER	passenger train	asserted; absent
unlearn	VizWiz	coffee beans	asserted; absent
unlearn	AMBER	beach ball	asserted; absent
learn	VizWiz	table	denied; present
learn	NoCaps	floor	denied; present
learn	AMBER	sky	denied; present
learn	VizWiz	floor (2nd img.)	denied; present
learn	COCO	truck	denied; present
learn	COCO	backpack	denied; present

The extended sample sharpens both regularities. Over-claims are uniformly *scene-schema completions*: a life jacket on a boat, a traffic light at an intersection, a winch on machinery, coffee beans near a coffee product, the model completes a scene schema rather than verifying visual evidence, the same co-occurrence mechanism DASH mines at web scale [4], here identified per instance and tagged with its direction. Misses skew toward *background and support surfaces* (floor, table, sky) and *partially occluded mid-ground objects* (truck, backpack), suggesting the no-bias stream teaches attention to unremarked regions rather than rare categories. That the two streams have such different visual anatomies is itself an argument for treating them with different losses.

L Probe-Sensitivity Analysis

Table 24 contrasts the two measurement regimes on the same base model and pool. Greedy decoding reports a moderate over-claim problem; the log-likelihood probe reveals that on adversarial absent objects the model's latent preference favors assertion in 80.9% of cases, for many queries the two answers are nearly equiprobable, and the sampled "no" conceals a marginal preference. The two regimes agree on the *direction* (both $s > 0$) but differ five-fold in magnitude. Since re-ranking, option scoring, and temperature sampling all consult the underlying likelihoods, we treat the latent probe as the deployment-relevant stress measure and greedy decoding as the user-visible one; both are reported wherever feasible.

Table 24: Greedy vs. latent (log-likelihood) measurement, Qwen2-VL-2B base model.

Regime / pool	FP rate	FN rate	s
Greedy, full 4,006-query pool	14.3%	6.5%	+0.386
Latent, 250 held-out adversarial	80.9%	0.7%	+0.978

M Gradient Analysis of the DUAL Objective

This appendix analyzes the three terms of Eq. (2) at the gradient level, explaining (i) why NPO-based unlearning is stable but weak in isolation, (ii) why the learn term in isolation collapses the model, and (iii) qualitatively how the controller of Eq. (3) balances the two.

M.1 Adaptive Damping of the Unlearn Term

Write $r = \log \frac{\pi_\theta(y^- \mid v,x)}{\pi_{\text{ref}}(y^- \mid v,x)}$ for the policy-to-reference log-ratio of the hallucinated assertion. The NPO term is $\ell_u(r) = -\frac{2}{\beta}\log\sigma(-\beta r)$, with

$$\frac{\partial \ell_u}{\partial r} = 2\,\sigma(\beta r) \in (0,2). \tag{5}$$

The gradient magnitude on the suppressed assertion is gated by $\sigma(\beta r)$: as unlearning progresses and $\pi_\theta(y^-)$ falls below the reference (r to $-\infty$), the gate closes exponentially. This is the collapse-slowing property inherited from NPO [39], and it also explains the near-null result of the unlearn-only arm in Table 1: from the very first step the term only ever *reduces* the assertion probability toward (and below) the reference, it contains no force that raises the probability of the *correct* answer, so a model whose assertion prior is strongly skewed retains its ranking between "Yes." and "No." almost everywhere (s: +0.978 to +0.977). Suppression alone therefore cannot rebalance the assertion prior.

M.2 Why the Learn Term Alone Collapses

The learn stream contains only records whose gold answer is `Yes.` (missed present objects). In isolation, its cross-entropy gradient $-\nabla_\theta \log \pi_\theta(\text{"Yes."} \mid v, x)$ raises the probability of the answer-initial token *Yes* for every training context. Because the LoRA update acts on shared attention projections, not on per-image circuits, a portion of this update is context-independent: it moves the language-tower's answer prior toward assertion in *all* presence-question contexts, including those where the correct answer is `No.` With no counter-pressure in the batch (no absent-object records at all), the context-independent component accumulates unopposed, and the model reaches the degenerate optimum of the stream, answering "yes" unconditionally (FP 100%, Table 1). The retain stream alone cannot prevent this, because retain records are dominated by the model's existing correct behavior and carry little gradient once fitted. The failure is thus structural, a direction-unbalanced batch *is* a biased objective, which is the gradient-level restatement of the paper's central claim.

M.3 The Controller as Proportional Balance Feedback

Let $\delta_u > 0$ denote the (average) reduction in the model's assertion margin caused by one unlearn step, and $\delta_l > 0$ the increase caused by one learn step, both measured on the probe distribution. Between consecutive probes, the expected drift of the signed score satisfies, to first order,

$$\mathbb{E}[\Delta s_t] \;\propto\; -\big(p_{\mathcal{U}}(t)\,\delta_u \;-\; q_{\mathcal{L}}(t)\,\delta_l\big), \tag{6}$$

where $q_{\mathcal{L}}$ is the learn-draw probability. The proportional rule $p_{\mathcal{U}} = \mathrm{clip}(0.5 + \gamma s_t, \cdot)$ makes the drift sign-opposing: yes-biased states ($s_t > 0$) increase unlearn pressure and vice versa, so $s = 0$ is the unique attractor within the clip region, and the dead-band $|s_t| \le \varepsilon$ converts the attractor into a stopping set. Because δ_u and δ_l are unknown and drift as the adapter trains, we make no quantitative convergence claim; the fixed-weight run of Table 1 ($p_{\mathcal{U}}$ constant, no stop) empirically demonstrates the failure mode the feedback removes, the state passes through $s = 0$ and continues to -0.657.

N Notation

Table 25: Notation used throughout the paper.

Symbol	Meaning
$\pi_\theta, \pi_{\mathrm{ref}}$	subject policy; frozen reference (adapter off)
v, x	image; text query
y^+, y^-	gold answer; hallucinated assertion ("Yes.")
$\mathcal{U}, \mathcal{L}, \mathcal{R}$	unlearn / learn / retain streams
FP, FN	over-claims; misses
s, s_t	behavioral signed score (Eq. 1); at probe t
λ_u, λ_r	unlearn / retain loss weights
β	NPO inverse-temperature
$p_{\mathcal{U}}(t)$	unlearn-draw probability set by Eq. (3)
K, γ, ε	probe period; controller gain; dead-band
$\mathcal{P}$	held-out balanced probe set

O Extended Benchmark Comparison

Table 26 situates the evaluation resources discussed in Section 2 along task format, primary target, and whether failure *direction* is an explicit output of the benchmark's protocol. Direction-awareness in the sense used by this paper, a signed quantity that can route an intervention, appears only in the suite we build on; BEAF's change-based metrics (stubbornness, indecision) are the nearest neighbors, characterizing answer instability under scene edits rather than a signed assertion bias.

P Behavioral Streams and the Judge-Side Adjective Partition

The training signal in this paper is behavioral (FP/FN membership). The judge-based diagnosis of the underlying suite instead extracts failure-describing *adjectives* from a judge LLM and maps them through a frozen 16-word partition. Table 27 reproduces the partition and aligns it with our streams: yes-bias adjectives describe exactly the behavior our unlearn stream suppresses, and no-bias adjectives the behavior our learn stream repairs; generic adjectives correspond to perception failures that both streams leave untouched (they surface in either stream depending on the direction the misperception happened to take). This alignment is what lets either diagnosis serve as the compass (Section 6): the streams identify *which* examples to treat, the adjectives characterize *what kind* of failure each represents, and the signed scores of the two diagnoses agree at scale (Table 6).

Table 26: Evaluation resources for VLM hallucination. "Dir." = does the protocol output an explicit failure-direction signal?

Resource	Venue	Format	Primary target	Dir.
CHAIR [24]	EMNLP'18	caption	object recall	–
POPE [16]	EMNLP'23	yes/no	object presence	△
NOPE [19]	ALVR'24	VQA	negative presence	–
AMBER [28]	arXiv'23	mixed	attr./relation	–
HallusionBench [8]	CVPR'24	VQA	illusion/language	–
MMHal-Bench [27]	ACL'24(F)	free-form	response quality	–
BEAF [33]	ECCV'24	yes/no×2	change consistency	△
HALLUCINOGEN [25]	UncNLP'25	implicit	yes-bias escalation	△
DASH [4]	ICCV'25	retrieval	systematic FPs	–
HalluCompass [2]	u.r.'26	bin.+free	signed direction	✓

Table 27: The frozen adjective partition of the underlying suite, aligned with DUAL-Compass streams.

Class	Adjectives	Stream
YesBias	overconfident, overinclusive, optimistic, careless, misinformed, biased	$\mathcal{U}$ (unlearn)
NoBias	overcautious, partial, incomplete, superficial, overlooked	$\mathcal{L}$ (learn)
Generic	misperceptive, misrecognized, misinterpreting, misleading, inaccurate	either

Q Dataset Licenses and Compute Budget

Licenses. All image bytes derive from the underlying suite's release and retain their original licenses: MS-COCO val2014 (CC-BY 4.0), AMBER (research-only redistribution), NoCaps validation (CC-BY-SA), VizWiz validation (CC-BY 4.0). Derived queries, stream files, adapters, and code will be released under CC-BY 4.0 (data) and MIT (code), matching the upstream suite. No new images, personal data, or human-subject data are collected; VizWiz images (taken by blind users) are used strictly under their released research terms.

Compute budget. Table 28 itemizes the GPU cost of every experiment in this paper. The full pilot, diagnosis, expansion, three ablation arms, closed loop, and utility audit for the primary model, fits within approximately twelve H100-hours; the cross-model replication approximately doubles this. For comparison, preference-based alignment pipelines report an order of magnitude more for the feedback stage alone [30].

R Case-Study Walkthrough

We trace one real record end-to-end to make the pipeline concrete.

1. Query. Image `AMBER_267.jpg` (an equestrian scene); query *"Is there a saddle in this image? Answer with only 'yes' or 'no'."*; gold label *no* (the annotation's `hallu_adv` list contains *saddle*: plausible near a horse, verified absent).

Table 28: Approximate compute budget (H100, shared-node wall-clock; unshared times in parentheses where measured).

Phase	GPU-hours
Diagnosis pass, 4k queries (2B)	1.7
Diagnosis pass, 8k queries, per model × 3	1.5–2.5 each
Ablation arm (150 steps + 2 evals) × 3	<1 each
Closed-loop run ($\leq$300 steps + 12 probes), per model	1–2 each
Utility audit (80 captions), per model	0.3 each
Primary-model pilot total	$\approx$ 12

2. Diagnosis. Greedy decoding yields *"Yes"*. The record enters the unlearn stream $\mathcal{U}$ with direction tag `yes_bias`, an auditable statement: *the model over-claimed a contextually plausible absent object.*

3. Training step. When drawn (probability $p_{\mathcal{U}}(t)$), the NPO term computes the summed log-likelihood of `Yes.` under the adapter and under the adapter-disabled reference; the gated gradient ($2\sigma(\beta r)$; Appendix M) lowers the assertion's likelihood ratio. No gradient ever targets `No.` on this image, suppression, not substitution, so related true assertions (*horse*, *fence*) are untouched except through shared parameters, which the retain stream anchors.

4. Post-training audit. After training, the record is re-scored. Under the DUAL arm of Table 1, records of this type flip to *No* in the large majority of held-out cases (FP 80.9% to 5.5%); the per-direction repair counts reported by the training script state exactly how many $\mathcal{U}$-type and $\mathcal{L}$-type failures were repaired, persisted, or newly introduced.

Contrast. The same record under the learn-only arm receives no suppression at all; moreover, the global assertion drift induced by the unbalanced learn stream (Appendix M) makes this over-claim *more* likely, the mechanism behind the 100% FP collapse.

S Practitioner's Checklist for Direction-Aware Repair

Distilled from Sections 5.2–5.9, for teams applying diagnosis-driven fine-tuning to a deployed VLM:

(1) **Diagnose at scale.** Compute the signed score on the full available probe pool, never on small per-cell subsets: below a few hundred errors the sign itself is unreliable (Table 6).
(2) **Match the probe to the deployment.** If your system re-ranks, scores options, or samples, diagnose with the log-likelihood probe; greedy-only diagnosis can understate the latent assertion prior five-fold (Appendix L).
(3) **Match the deployment's source distribution.** Direction can flip between curated and in-the-wild imagery (Appendix D); build streams from the distribution you serve.
(4) **Do not train on a single direction.** A learn-only batch is a biased objective (Appendix M); always co-administer the opposite pressure and a retain anchor, even when one stream is small.
(5) **Adapt during training rather than fixing a schedule.** Fixed mixing overshoots; re-diagnose during training and stop within the dead-band (Section 4.3).
(6) **Audit both directions after training.** Report FP and FN movement together; an FP reduction obtained at the cost of a large FN increase substitutes one deployment failure for another rather than improving the model.
(7) **Re-run safety evaluations.** Any behavioral fine-tune can shift unrelated capabilities; utility audits (Appendix F) bound generative regressions but do not replace full evaluation.

T Threats to Validity

Table 29: Limitations and the evidence that bounds each (referenced from Section 6).

Limitation	Status / mitigating evidence
Object-presence scope	Free-form generation measured via utility audit (§5.5); span-level extension designed (App. U)
External transfer (scope)	Claims are on-distribution; cross-distribution transfer to DASH-B hard negatives is unsolved by in-suite or DASH-train supervision (App. X) — an explicit boundary, not a claim
Judge dependence	Training uses behavioral diagnosis (no judge); adjective judge agrees in sign at scale (Table 6)
Seed sensitivity	Main arms reported over 3 seeds (Table 1: unlearn 61.2±3.8, learn 56.0±0.0, DUAL fixed 82.4±4.6); the closed loop removes the fixed-weight seed variance; some appendix ablations remain single-seed
Probe distribution	Suite-drawn probe can under-represent adversarial bias (2B); documented and quantified (§5.9)
Image resolution	512-px training; native-resolution re-evaluation shifts accuracy <1pp (App. I)

Internal. All ablation arms share hyperparameters, data pool, and step budget, so Table 1 differences are attributable to the objective; the main arms are reported over three seeds (the ordering is seed-invariant), though some appendix ablations remain single-seed and step-budget interactions (e.g., whether unlearn-only would move at 10× steps) are unexplored. The learn-only collapse is an extreme point: gentler learn-dominant mixtures would collapse more slowly, and we have not mapped the full mixture–outcome surface. **Construct.** The behavioral signed score treats all over-claims and misses as exchangeable within their direction; adjective-level diagnosis (Appendix P) distinguishes failure kinds within a direction, and per-kind repair analysis is future work. The automatic caption metric is substring-based and conservative (Appendix F); the human audit (Section 5.8) bounds its error and independently confirms the quality gain (89% of captions judged hallucination-free, 85% preferred over the baselines). **External.** Evidence covers 2–7B open VLMs on object presence over four public image sources; extrapolation to frontier-scale models, attribute/relational hallucination, and non-English queries is unsupported by our data. Streams and evaluation come from one suite's disjoint splits; cross-benchmark transfer (DASH-B, BEAF) is planned. **Statistical.** Held-out evaluation uses 250 queries; the headline effects (75.4pp FP reduction, 44pp collapse gap between arms) are far outside binomial noise at this n (±6pp at 95%), but small deltas (e.g., unlearn-only's +0.8pp) are within it and we do not interpret them.

U Extension Design: Signed Span-Level Unlearning

The binary-presence instantiation of DUAL-Compass is deliberately minimal; this appendix sketches the free-form extension, which reuses every component.

Stream construction. Run the subject model's free-form descriptions through the suite's judge protocol (or a grounding detector) to label, per generated span, whether it asserts an absent object (yes-bias span), omits a present prominent object (no-bias omission, defined at the response level), or is faithful. This mirrors EFUF's CLIP-threshold span selection [30] but with a signed, explainable label per span.

Objective. Eq. (2) transfers with token masks: the NPO term scores only yes-bias span tokens (suppressing the assertion in context); the learn term supervises corrected responses that include the omitted objects (obtainable from annotations without human writing, by template or judge rewrite); the retain term anchors faithful spans and overall fluency, subsuming EFUF's sentence-level loss as its undirected special case.

Control. The probe becomes a small free-form set scored by the judge; the signed score is the normalized difference between yes-bias span rate and omission rate, and Eq. (3) applies unchanged. The open risks are judge cost inside the loop (mitigable by probing every larger K, or by the behavioral binary probe as a cheap surrogate given the sign agreement of Table 6) and omission supervision quality; both are measurable with the instruments of Appendices F and H.

V Extended Discussion of Evaluation Resources

This appendix expands the capsule descriptions of Section 2 and Table 26, focusing on what each resource operationalizes and what it leaves open for a training-side, direction-aware method.

CHAIR [24] scores hallucinated objects in captions against COCO ground truth, establishing object hallucination as measurable but conflating all error kinds into caption-level rates; no probing of the model's assertion prior is possible.

POPE [16] isolates the assertion prior with balanced yes/no probes and popularized the random/popular/adversarial negative hierarchy our splits inherit. Its yes-ratio is the closest ancestor of a direction signal, but as an unsigned aggregate over a balanced pool it cancels opposing errors, the starting observation of the suite we build on.

NOPE [19] stresses negative-presence questions specifically, showing models struggle to say "nothing"; it measures one direction's failure without contrasting it against the other.

AMBER [28] contributes LLM-free multi-dimensional evaluation (existence, attribute, relation) and the curated image pool our annotations extend; its metrics remain unsigned rates per dimension.

HallusionBench [8] disentangles language-prior failures from visual-illusion failures with paired image edits, a causal probe of *why* a model errs, complementary to our *which direction* framing.

MMHal-Bench [27] evaluates free-form response quality with GPT-4 judging, foreshadowing the judge-dependence trade-offs we discuss in Section 6.

BEAF [33] edits the scene itself and scores answer changes, yielding change-consistency metrics (true understanding, stubbornness, indecision). Stubbornness under object removal is behaviorally adjacent to yes-bias, and BEAF's manipulated pairs would make a strong external test of whether DUAL-trained models update answers when the evidence changes.

HALLUCINOGEN [25] demonstrates that implicit prompts escalate yes-bias predictably along a difficulty gradient, evidence that assertion bias is a graded, manipulable quantity, which our controller exploits in training rather than measurement.

DASH [4] scales false-positive discovery to the open world with retrieval-guided mining; its clusters constitute, in our terms, a web-scale unlearn stream that lacks direction-balanced counterparts. Combining DASH-mined $\mathcal{U}$ with annotation-derived $\mathcal{L}$ is a natural scaling path for DUAL-Compass.

HalluCompass [2] supplies the signed diagnosis, the frozen adjective partition (Appendix P), the source-controlled image pool, and the finding that misrouted prompt interventions harm, the evaluation-side half of the program this paper completes on the training side.

W Full Results Table

Table 30: Findings at a glance (Qwen2-VL-2B held-out adversarial unless noted). Each row is a claim the experiments establish.

Claim	Evidence
Simultaneity is required	DUAL 86.0% vs. unlearn-only 64.8%, learn-only 56.0% (collapse)
Balanced SFT is not enough	60.3%±14.4 over 3 seeds; POPE-compat 59.3%
Fixed weights are fragile	44–86% across orders/models; collapse on 3B, 7B
Control makes it reliable	86.4 / 88.8 / 84.8% on 2B / 3B / 7B; 0/9 collapses over 3 seeds (86.5±0.6 / 87.3±2.2 / 84.3±0.6)
Loss choice is second-order	NPO and GA both collapse fixed, both stable under control
Beats EFUF-style unlearning	matches acc. but overshoots to no-bias ($s = -0.67$ vs. +0.53)
Utility is preserved	caption hallucination unchanged within noise; 0 degenerate
Humans confirm it	diagnosis 90% human-valid; captions preferred 85% over prompt/EFUF
Diagnosis needs scale	small-sample sign inverts on 5/6 models
Claims are on-distribution	cross-distribution (DASH-B) transfer is out of scope
External POPE (COCO) confirms	direction is distribution-dependent: suite-loop repairs 7B (+.41→ + .15); COCO-diagnosed loop lifts 2B 73.0→91.3 (App. X.2)

Table 31 consolidates every training run reported in the paper on the common held-out adversarial protocol, for auditability. Fixed-weight rows vary only in data order/seed and step budget; their spread is the fragility the controller removes.

Table 31: All 2B training runs (held-out adversarial). s = **signed score after training; base is** +0.98 / 64.0%.

Method	Config	Acc.	s
Unlearn only (NPO)	150 st.	64.8%	+0.98
Learn only (SFT)	150 st.	56.0%	+1.00
Balanced SFT	seed 0	62.8%	−0.96
Balanced SFT	seed 1	44.8%	−1.00
Balanced SFT	seed 2	73.2%	−0.70
Fixed DUAL (NPO)	order A	86.0%	−0.66
Fixed DUAL (NPO)	seed 1	77.2%	−0.90
Fixed DUAL (NPO)	seed 2	84.0%	−0.55
Fixed DUAL (NPO)	450 st.	71.6%	−0.75
Fixed GA (0.3)	450 st.	74.0%	−0.97
Fixed GA (1.0)	150 st.	44.0%	−1.00
Fixed NPO (λ_u=3)	150 st.	44.4%	−1.00
Loop + ckpt. (NPO)	damped	**86.4%**	+0.53
Loop + ckpt. (GA 0.3)	damped	**86.8%**	+0.35
Loop + ckpt. (GA 1.0)	damped	**86.8%**	+0.21
Loop, undamped (NPO)	γ=0.6	44.0%	−1.00

X Cross-Distribution External Evaluation

X.1 DASH-B: Stream Augmentation and Scaling

The main text reports that supervision built only from the suite's curated negatives does not transfer to DASH-B's web-mined systematic false positives, and that mixing a DASH-diagnosed stream moves the external true-negative rate (TNR) in the right direction. This appendix details the protocol and the scaling behavior.

Protocol and eval integrity. DASH-B provides 2,682 examples. We reserve the first 1,000 (under a fixed seed-0 shuffle) as a held-out evaluation set used for *every* DASH-B number in the paper, and use only the remaining rows for stream construction, the evaluation set is never trained on. Diagnosing the 2B model on the train rows yields a strongly yes-biased stream ($s_{\text{beh}} = +0.90$; 430 over-claims vs. 22 misses), quantifying why DASH-B is hard relative to the suite (+0.35).

Why a naive mix under-transfers. In an even mix, the DASH over-claim records (430) are diluted by the suite over-claim records (566), and, more importantly, the controller's balance probe is drawn from the suite streams, so it never *sees* DASH negatives and cannot steer toward DASH TNR. The augmented run therefore (i) over-samples the DASH over-claim stream so it dominates the unlearn pool, and (ii) admits DASH negatives into the probe, so the signed score the controller minimizes reflects DASH balance.

Scaling result: the stream carries the signal, but naive scaling trades one direction for the other. Oversampling the DASH over-claim stream four-fold and increasing the budget confirms the stream is informative, a *fixed-weight* DASH-heavy run drives DASH-B TNR from the base 53.6% to 100%, but does so by collapsing to universal denial (TPR 8.9%, yes-ratio 0.04): it learns to answer "no" to DASH-style questions, not to see the images. The corresponding *closed-loop* run, whose controller balances a probe still dominated by the suite distribution, over-corrects on the suite (s=−0.89) while remaining yes-biased on DASH-B (TNR 5.6%). The same adapter is thus no-biased on one distribution and yes-biased on the other, direct evidence that a single scalar controller cannot jointly balance two distributions whose difficulty differs.

Distribution-matched control does not close the gap either. Balancing the controller on a DASH-representative probe (balanced DASH positives/negatives, disjoint from the 1,000-query evaluation set) reaches probe balance during training and preserves in-suite performance (held-out adversarial 87.2%, POPE-compat 82.0%), but the held-out DASH-B TNR remains low (27.2%, TPR 98.2%): the model is balanced in log-likelihood on the DASH probe yet greedy-decodes yes-biased on the DASH evaluation, the same latent-vs-greedy gap documented in Appendix L, now across distributions. Summarizing the sweep: no intervention we tried achieves a genuine (non-collapsed) improvement in external DASH-B TNR from in-suite or DASH-train supervision. We report this as an honest negative result that bounds the method's scope. Closing it plausibly requires (i) a controller that optimizes the greedy-decoding objective the benchmark scores, and (ii) supervision at DASH's scale rather than its 1,682-row train split; both are beyond this paper's scope but consistent with its framework, since each distribution contributes its own diagnosed stream and probe.

X.2 Distribution-Matched (COCO) Repair Protocol

This appendix documents the constructive experiment of Section 5.9 (Table 32). The official POPE-COCO pool of 500 images is split deterministically (sorted by filename) into the first 300 for diagnosis and training and the disjoint last 200 for held-out evaluation, *no training image appears at evaluation.* The subject model is diagnosed on the 300 training images' presence queries (greedy), producing direction streams; because both Qwen models are no-biased on COCO the streams are learn-heavy (2B: 11 unlearn / 1,287 learn / 4,102 retain; 3B: 174 / 498 / 4,728). The compass loop then trains with a COCO-drawn balanced probe and best-balance checkpointing (same hyperparameters as the main loop, Table 21). Evaluation is greedy on the held-out 200 images (400 queries per POPE split). Base signed scores on this 200-image split (−0.97 for 2B, −0.46 for 3B) are milder than the full-500 scores in Table 7 (−0.99, −0.62) because the subset is smaller and differently sampled.

Table 32: Distribution-matched (COCO-diagnosed) repair, held-out 200-image POPE-COCO split (greedy). Diagnosing on the deployment distribution reverses the mismatched regression on the flagship 2B; 3B over-corrects past balance.

Model	base acc	COCO-loop acc	base s	COCO-loop s
Qwen2-VL-2B	**73.0%**	**91.3%**	**−0.97**	**−0.21**
Qwen2.5-VL-3B	87.7%	84.2%	−0.46	+0.47

Y Reproducibility Checklist

- **Models.** All subject models are public: `Qwen2-VL-2B-Instruct`, `Qwen2.5-VL-3B-Instruct`, `llava-1.5-7b-hf` (Unsloth mirrors).
- **Data.** All images, annotations, and queries are from the public HalluCompass release; DASH-B from its public release. The direction-balanced expansion is rule-based (App. C) and deterministic under seed 0.
- **Code.** Five scripts (App. J) reproduce diagnosis, expansion, fixed/loop training, and all evaluations; released under MIT.
- **Determinism.** Evaluation is sampling-free (log-likelihood comparison). Training uses seed 0 unless a seed is stated; we note the unsloth model-load reseeding pitfall and re-seed immediately before stream shuffling.
- **Hardware / cost.** Single H100 per run; full primary-model pilot ≈12 GPU-hours (App. Q).
- **Hyperparameters.** Complete in Table 21; identical across ablation arms.
- **Metrics.** FP/FN rate, accuracy, signed score (Eq. 1) on held-out adversarial; standard greedy POPE-compat (acc/P/R/F1/yes-ratio); DASH-B TNR/TPR; caption hallucination/length/degeneration for utility.
- **Statistics.** Multi-seed for decisive comparisons; headline effects far exceed the ±2.7pp binomial band at the full-split n=1,341.
- **Known limitations.** Enumerated in Table 29 and App. T.

Z Broader Impact

DUAL-Compass targets a deployment-safety failure (fabricated visual assertions) while explicitly protecting the complementary utility failure (missed objects), and it makes the intervention auditable at the level of failure direction. Positive impact: teams can repair a diagnosed bias with a low-cost, transparent procedure rather than an opaque uniform pressure, and can verify *which* direction they moved. Risks and mitigations: (i) any behavioral fine-tune can shift unrelated capabilities, we recommend re-running full safety evaluations post-training and provide a utility audit as a first screen; (ii) a controller optimized to a mis-specified probe distribution can balance the wrong thing, we document the probe-distribution effect and recommend diagnosing on the deployment distribution; (iii) the method reduces but does not eliminate hallucination, and should not be represented as a safety guarantee. No new models, scraped data, or personal data are released; all artifacts derive from public research datasets.